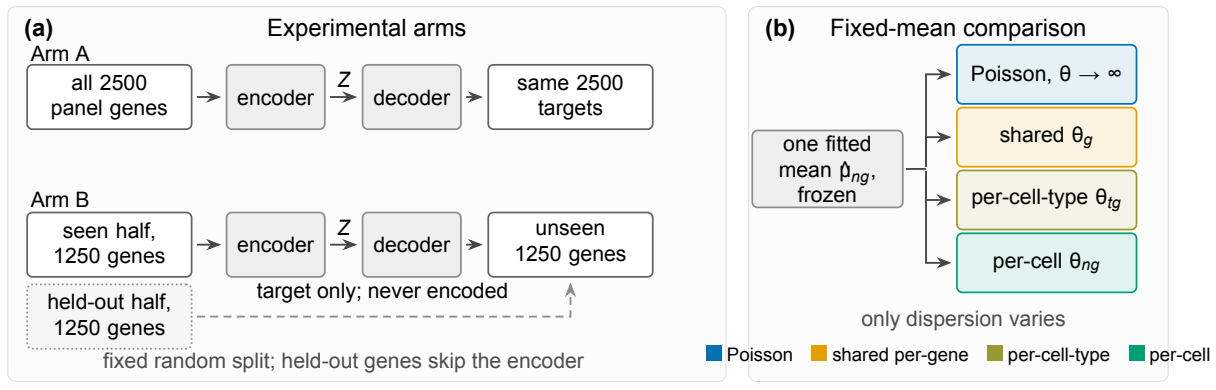
(a)
Experimental arms
Arm A
all 2500 panel genes
encoder
Z
decoder
same 2500 targets
Arm B
seen half, 1250 genes
encoder
Z
decoder
unseen 1250 genes
held-out half, 1250 genes
target only; never encoded
fixed random split; held-out genes skip the encoder
(b)
Fixed-mean comparison
one fitted mean $\hat{\mu}_{ng}$, frozen
Poisson, θ → ∞
shared θ_g
per-cell-type θ_{tg}
per-cell θ_{ng}
only dispersion varies
Poisson
shared per-gene
per-cell-type
per-cell